

Depth-Regularized Sparse Multiple Instance Learning for Mid-Range Tree Species Classification under Background-Dominant Conditions

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ABSTRACT Tree species classification from ground-level images is important for urban forestry, orchard management, and ecological monitoring. Existing recognition systems often assume close-up views of leaves, flowers, fruits, or bark, whereas practical field images are frequently captured from mid-range distances. In this setting, discriminative evidence from trunks, branches, and bark occupies only a small part of the image, while buildings, roads, sky, signs, vehicles, and neighboring trees dominate the background. This creates a background-dominant weakly supervised recognition problem: image-level labels are available, but the informative regions are sparse and not manually annotated. This paper proposes a depth-regularized sparse multiple instance learning framework for mid-range tree species classification. Images are converted into bags of local patches, trunk-biased sampling selects structurally informative patches using gradient-based scores, and attention-based multiple instance learning aggregates the selected instances. Depth-guided augmentation is used only during training to preserve foreground tree structures while perturbing background regions, so inference requires RGB images only. Experiments on FruitsPark and UrbanStreetTree compare image-level baselines, sparse MIL, and sparse depth-augmented MIL under a four-seed protocol. In the fixed-budget backbone comparison on UrbanStreetTree, ViT DepthAug achieves 95.10% accuracy and 94.27% macro- F_1 , improving over the corresponding ViT GAP baseline by 10.49 and 12.50 percentage points, respectively. A separate DINOv2 ratio sweep further shows that near-peak performance can be retained with substantially fewer patches: on UrbanStreetTree, a 0.4 bag ratio achieves 95.98% accuracy, within 0.35 percentage points of the best result, while using 163.3 rather than 367.2 instances on average. These results indicate that depth-regularized sparse MIL is an effective and patch-efficient strategy for tree classification under background-dominant mid-range imaging conditions.

INDEX TERMS Attention-based multiple instance learning, background bias, depth-guided augmentation, sparse patch selection, tree species classification, urban forestry, Vision Transformer.

I. INTRODUCTION

Automatic tree species classification supports urban green-space management, biodiversity surveys, orchard monitoring, and smart-city environmental inventories. Mobile and ground-level imaging is attractive because it is inexpensive and easy to deploy, but most successful plant and tree recognition pipelines have relied on close-up observations of leaves, flowers, fruits, or bark [1]–[4]. Such assumptions do not always hold in field surveys. A user may capture an entire tree from several meters away, or an urban inventory

system may observe a street tree from a pedestrian or vehicle viewpoint. In these mid-range images, the tree is only partly visible and the background often occupies most of the frame.

This study focuses on the specific difficulty created by this imaging condition. The discriminative signal for species recognition is concentrated in trunks, branches, bark patterns, and local tree structures, but these regions are not annotated. At the same time, the image contains a large amount of scene context such as buildings, roads, sky, vehicles, signs, ground texture, and neighboring vegetation. Modern classifiers can



FIGURE 1. Examples of tree images captured at different distances. Mid-range observations contain larger background regions and weaker local bark texture than close-up observations.

exploit such context as a shortcut. Background dependence has been quantified in general object recognition [8] and in plant-disease datasets [9], where label-correlated background artifacts can be sufficient for non-trivial classification. In mid-range tree images, this risk is amplified because the object evidence is sparse and the background area is large.

Close-up bark classification provides a second useful but incomplete precedent. BarkNet 1.0 and CentralBark show that bark is a strong year-round cue for tree recognition [3], [4]. However, these datasets are dominated by close-range bark imagery. When a trunk is observed from a distance, the effective resolution of bark texture decreases, and the same apparent trunk width in the image may correspond to very different physical scales. Thus, mid-range tree species classification requires a method that can preserve local evidence without depending on manual trunk crops and can combine weak local cues with broader structural information.

We formulate this setting as a background-dominant multiple instance learning (MIL) problem. An image is represented as a bag of patches, only the image-level species label is available, and the classifier must infer which instances are informative. Attention-based MIL (ABMIL) provides a differentiable pooling mechanism that assigns weights to instances and yields patch-level diagnostics [15]. However, full-patch MIL is computationally expensive when high-resolution images are processed densely, and recent sparse MIL studies have shown that instance selection, masking, and pruning are important design axes rather than mere implementation details [19]–[23]. Naive sparse sampling can discard useful trunk or branch patches. We therefore introduce trunk-biased sampling (TBS), which selects a sparse subset of candidate patches by combining sliding-window coverage with gradient-based structural scores. To reduce background shortcuts during training, we further use depth-guided augmentation (DepthAug), which preserves depth-derived foreground structures while perturbing background regions. Depth is used only for data construction or training-time augmentation; the final model uses RGB images only at inference.

The contributions of this paper are as follows:

- We formulate ground-level mid-range tree species classification as a background-dominant weakly supervised patch aggregation problem.
- We propose a sparse MIL framework that combines

trunk-biased patch selection, Vision Transformer-based feature extraction, and ABMIL aggregation.

- We incorporate depth-guided augmentation as a training-only regularizer for suppressing background shortcuts without requiring depth at inference.
- We evaluate image-level baselines, sparse MIL, and sparse DepthAug MIL under a unified four-seed protocol on FruitsPark and UrbanStreetTree.
- We quantify the accuracy–patch-budget Pareto frontier and show that near-peak accuracy can be maintained with substantially fewer instances, especially on UrbanStreetTree.

II. RELATED WORK

A. TREE SPECIES CLASSIFICATION

Tree species recognition has been studied using leaves, flowers, fruits, bark, whole trees, aerial imagery, satellite imagery, and LiDAR. LeafSnap and Pl@ntNet popularized image-based plant identification with close-up organ observations [1], [2]. Bark-based recognition is attractive because bark can be observed throughout the year. BarkNet 1.0 introduced a large bark dataset and reported strong performance with CNN-based classifiers [3]; CentralBark further expanded standardized bark imagery for deep learning-based species recognition [4]. These studies demonstrate the value of bark texture, but their close-range assumptions differ from our mid-range setting, where the trunk is small and background interference is substantial.

Urban tree datasets have also expanded. UrbanStreetTree provides a comprehensive public dataset for image classification and instance segmentation across street-tree organs, including branches, and contains 41,467 high-resolution classification images across 50 species [5]. Recent large-scale benchmarks such as StreetTree further show that street-level tree species classification is becoming an active research area [6]. Other studies use multisource observations such as UAV RGB imagery and LiDAR to exploit canopy color, texture, height, and laser intensity cues for urban tree classification [7]. These works show the value of structural sensing, but they generally rely on remote-sensing views or additional input modalities. In contrast, our setting is ground-level mid-range RGB classification, where target trunk and branch evidence is sparse, no trunk-level annotation is available, and the deployed model should not require depth or LiDAR input.

B. BACKGROUND BIAS AND MID-RANGE RECOGNITION

Background bias is a major reliability issue in visual recognition. Xiao et al. showed that background-only information can support non-trivial ImageNet-9 classification and that adversarially chosen backgrounds can induce incorrect predictions even when the foreground is correctly classified [8]. Noyan reported that background pixels in PlantVillage contain label-correlated artifacts, allowing a classifier using only a few background pixels to exceed random guessing by a large margin [9]. These findings are directly relevant to

mid-range tree recognition because background pixels often dominate the image.

Mid-range imaging also reduces the discriminative value of local texture. CNNs are known to rely strongly on texture cues, while shape-biased learning can improve robustness [10]. For tree species classification, this implies that close-up bark texture alone is insufficient; a model must combine local texture remnants with trunk and branch structure while avoiding scene context shortcuts.

C. WEAKLY SUPERVISED INSTANCE ATTENTION FOR TREE RECOGNITION

Weak supervision is relevant to tree recognition because exhaustive local annotations are expensive. Aygunes et al. proposed weakly supervised instance attention for multisource fine-grained object recognition and applied it to tree species classification under location uncertainty, combining information from sources such as RGB, multispectral, and LiDAR data [11]. This work is an important precedent because it shows that instance attention can help tree recognition when object evidence is weakly localized. However, it is an aerial and remote-sensing study: the instances correspond to uncertain object locations in overhead multisource imagery, and structural information can be supplied by external modalities such as LiDAR. Our problem differs in viewpoint, evidence distribution, and deployment constraint. We consider ground-level mid-range RGB images rather than aerial multisource observations, and the useful evidence is concentrated in sparse trunk, branch, and bark patches surrounded by scene background. Depth is not used as an inference-time modality; it is used only to regularize training through background perturbation.

D. MIL, ABMIL, AND WEAK LOCAL EVIDENCE

MIL represents a sample as a bag of instances with only a bag-level label [12], [13]. This formulation directly matches the present annotation regime: each image has a species label, but the informative trunk, branch, and bark patches are not annotated. Deep MIL extended this idea to image classification and auto-annotation [14]. ABMIL introduced learnable permutation-invariant attention pooling for instance aggregation and provides instance weights that can be inspected as patch-level diagnostics [15]. We therefore use ABMIL as a standard and compact weakly supervised patch aggregator, rather than as the main novelty of the paper.

Weakly supervised whole-slide image analysis has further developed this line of work. CLAM uses attention and clustering constraints to identify representative high-value regions from slide-level labels [16], DSMIL models instance- and bag-level evidence with a dual-stream architecture [17], and TransMIL models correlations among instances with a Transformer architecture [18]. These works are conceptually related because they infer discriminative local regions from bag-level supervision. At the same time, many prominent MIL applications operate on large images whose instances are drawn from a relatively consistent domain, such as tissue

or other repeated local patterns. Ground-level natural images are less controlled: object scale, viewpoint, occlusion, lighting, and background content vary substantially, and label-correlated background regions may be more visually dominant than the target evidence. Applying MIL aggregation to mid-range natural tree images is therefore not a direct transfer of pathology-style MIL. In this setting, informative trunk and branch patches can be a small minority among visually strong background regions, which motivates explicit sparse patch selection and background-regularized training on top of ABMIL.

Vision Transformers (ViT) process images as patch tokens and can model long-range relationships [24]. Swin Transformer introduces hierarchical shifted-window attention [26]. Self-supervised vision backbones such as DINO, DINOv2, and DINOv3 learn general visual representations from large unlabeled corpora [27]–[29]. We evaluate such backbones in both image-level and MIL settings, focusing on whether they help under background-dominant sparse aggregation.

E. SPARSE AND EFFICIENT INSTANCE SELECTION

Recent MIL literature has increasingly questioned whether all extracted instances should be processed by the aggregator. PAMIL introduces dynamic policy-driven instance sampling for whole-slide classification and explicitly treats sampling as part of the learning problem [19]. MHIM-MIL masks salient easy instances to encourage discovery of harder useful instances [20]. ACMIL analyzes attention concentration and shows that a tiny number of instances can dominate attention, motivating mechanisms that suppress over-concentration [21]. Key Patches Are All You Need reports that using only a subset of regions can preserve in-domain diagnostic performance and improve robustness to shifts [22]. HDMIL further targets fast and accurate gigapixel classification by eliminating irrelevant patches [23].

These studies are mostly from computational pathology, not tree species recognition. Nevertheless, they are methodologically important because they address the same weak-label, many-instance, high-redundancy regime. Our contribution differs by moving sparse MIL to mid-range tree images and by using an application-specific trunk prior and training-only depth augmentation rather than learned pathology-specific relevance modules.

F. DEPTH AS TRAINING-TIME PRIVILEGED INFORMATION

Monocular depth estimation models such as MiDaS and Depth Anything V2 infer scene structure from RGB images [30], [31]. Many RGB-D recognition, detection, and segmentation systems use depth as an input modality at inference. A different line of work uses depth only during training and keeps RGB-only inference. This setting is related to learning using privileged information (LUPI), where additional information is available to the learner during training but not at test time [32]. In the present study, RGB images are the ordinary

input and pseudo-depth maps act as training-time privileged signals.

Training-only depth has been explored in several vision tasks. Zhao et al. asked whether depth is really necessary for salient object detection and proposed depth-aware regularization that uses depth during training while relying only on RGB at inference [33]. Fang et al. proposed depth removal distillation for RGB-D semantic segmentation, transferring knowledge from an RGB-D teacher to an RGB-only student [34]. Depth cues have also been incorporated into weakly supervised vision. Gungor and Kovashka used hallucinated monocular depth to improve MIL-based weakly supervised object detection through depth fusion and depth priors for proposal refinement [35]. DepthG uses depth-guided feature correlation and sampling for unsupervised semantic segmentation while not necessarily requiring depth at inference [36]. These studies show that predicted depth can provide useful structural information without requiring a depth sensor at deployment.

However, these studies do not establish whether training-only depth is effective for our setting. Salient object detection and semantic segmentation optimize dense foreground or pixel labels, depth removal distillation transfers an RGB-D segmentation teacher, and depth-assisted WSOD validates depth priors for proposal selection and pseudo-box refinement. Our task is species classification from ground-level natural images, where the classifier must aggregate sparse local tree evidence while resisting background shortcuts. We do not use LiDAR or depth as an input modality at inference, and we do not refine object proposals or pseudo boxes. Instead, pseudo-depth is used during training to preserve likely foreground tree structures while perturbing background regions in a sparse patch-based MIL classifier. Thus, although existing work supports the general usefulness of MIL aggregation and training-time depth signals, their effectiveness under ground-level mid-range tree classification remains insufficiently verified. This gap motivates evaluating depth-guided background regularization together with sparse MIL in the present task.

III. METHOD

A. OVERVIEW

The proposed framework converts a mid-range tree image into a sparse bag of structurally informative patches, extracts patch features with a visual backbone, and aggregates them with ABMIL. Depth-guided augmentation is applied during training to reduce dependence on background context. Fig. 2 illustrates the overall process.

B. IMAGE-TO-BAG CONSTRUCTION

Given an input image x , candidate patches are generated with a sliding window on the original image resolution. Let $\mathcal{C}(x) = \{q_1, \dots, q_M\}$ denote the candidate patch set. A sparse bag $\mathcal{B}(x)$ is selected from $\mathcal{C}(x)$ according to a bag ratio $\rho \in (0, 1]$, so that approximately $K = \text{round}(\rho M)$ patches are processed by the MIL model. Unlike a fixed patch

count, the ratio-based definition preserves a comparable fraction of available local evidence across images with different effective resolutions.

C. TRUNK-BIASED SAMPLING

TBS prioritizes candidate patches that are likely to contain high-frequency tree structures such as bark edges, trunk boundaries, and branch contours. Candidate patches are extracted with a 224-pixel window and a 160-pixel stride; border regions outside the sliding grid are not forcibly added. For each candidate patch q , a Sobel score is computed as

$$S_{\text{sobel}}(q) = \frac{1}{|q|} \sum_{u \in q} \|\nabla I(u)\|_2, \quad (1)$$

where I is the image intensity and u indexes pixels in the patch. Candidate patches are ranked by S_{sobel} . A trunk-bias parameter β controls the fraction of the selected bag that is deterministically filled from the highest-scoring candidates. The remaining instances are filled from lower-ranked candidates using a seed-controlled shuffle. In the reported sliding-window TBS setting, we use $\beta = 0.5$.

This selection is intentionally simple and deterministic. It is not intended to segment the trunk exactly; rather, it shifts the sparse bag toward structurally informative regions while retaining enough diversity for ABMIL to learn which patches are useful.

D. PATCH FEATURE EXTRACTION AND ABMIL

Each selected patch $p_i \in \mathcal{B}(x)$ is encoded by a backbone feature extractor $\phi(\cdot)$:

$$z_i = \phi(p_i) \in \mathbb{R}^d, \quad i = 1, \dots, K. \quad (2)$$

The ABMIL attention score is

$$e_i = w^\top \tanh(Wz_i + b), \quad (3)$$

and the normalized attention weight is

$$a_i = \frac{\exp(e_i)}{\sum_{j=1}^K \exp(e_j)}. \quad (4)$$

The bag representation is computed as

$$s = \sum_{i=1}^K a_i z_i, \quad (5)$$

and class probabilities are produced by a linear classifier followed by softmax:

$$p(y = c | x) = \text{softmax}(W_c s + b_c). \quad (6)$$

Training minimizes cross-entropy with a small penalty on the concentration of the normalized attention distribution:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_a \|a\|_2^2. \quad (7)$$

The effective sparsity in this work comes primarily from sparse instance selection by TBS rather than from an ℓ_1 penalty on softmax-normalized attention, whose ℓ_1 norm is constant. In the main protocol, $\lambda_a = 5 \times 10^{-4}$ and entropy regularization is disabled.

Depth-regularized sparse MIL pipeline

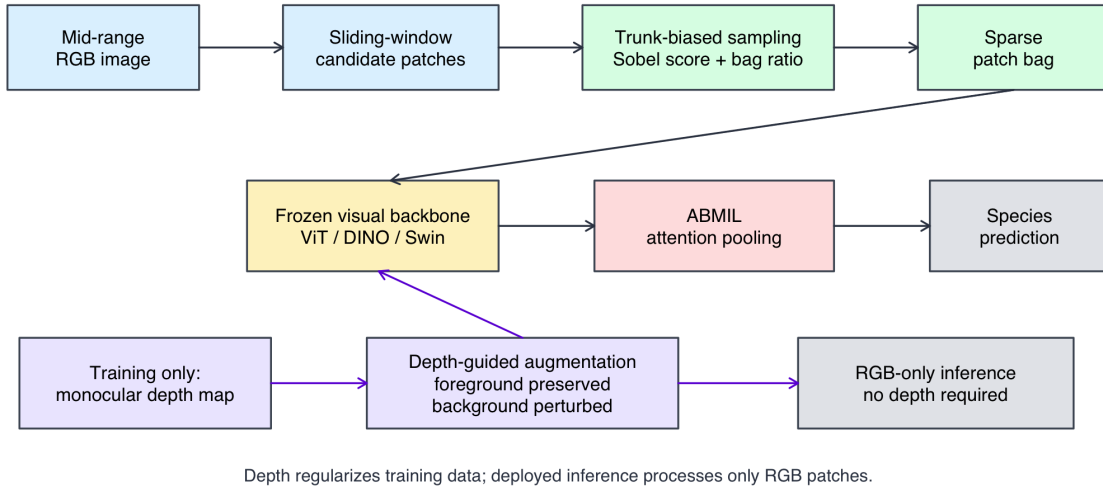


FIGURE 2. Overview of the sparse MIL framework. The current main line uses RGB patches at inference, trunk-biased sparse sampling, ABMIL aggregation, and training-only depth-guided augmentation.

E. DEPTH-GUIDED AUGMENTATION

DepthAug uses pseudo-depth maps estimated from RGB images to construct training images in which foreground tree structures are preserved and background regions are perturbed. The purpose is not to add depth as an inference-time modality, but to weaken label-correlated background shortcuts during learning.

Let $D(x)$ be a normalized relative depth map and $m(x)$ a foreground mask derived from $D(x)$ by thresholding and morphological smoothing. A depth-augmented training image is formed as

$$\tilde{x} = m(x) \odot x + (1 - m(x)) \odot T(x), \quad (8)$$

where $T(\cdot)$ applies background perturbations such as blur, color transformation, or compression. Validation and test images are not augmented in this way. At inference, the model receives only RGB images and uses the same TBS and ABMIL pipeline.

IV. EXPERIMENTAL SETUP

A. DATASETS

We evaluate on two datasets. FruitsPark is an in-house dataset collected at a public fruit park in Japan using mobile devices. It contains 790 original images from 10 tree classes and uses a train/validation split; because no independent test set is available, the validation split is used as the final evaluation split. UrbanStreetTree is a public street-tree dataset collected in Chinese cities [5]. We use the branch subset prepared for this study, containing 1,485 images from 13 classes with

TABLE 1. Dataset summary used in this study.

Item	FruitsPark	UrbanStreetTree
Classes	10	13
Original images	790	1,485
Train / val / test	580 / 210 / 0	1,193 / 149 / 143
Augmented train images	2,273	4,674
Final evaluation split	Validation	Test

train, validation, and test splits. Representative images are shown in Figs. 3 and 4.

B. COMPARED MODELS

The overall comparison includes three groups. First, image-level GAP baselines use ResNet18 [25], ViT [24], DINO/DINOv2/DINOv3 ViT backbones [27]–[29], and Swin Transformer [26]. These baselines use trainable backbones. Second, sparse MIL models use frozen backbones and TBS with $\rho = 0.3$. Third, sparse DepthAug models use the same sparse MIL setting but train on depth-augmented datasets. The arrows in Table 2 compare the reported image-level and MIL protocols within the same backbone family; they should be interpreted as protocol-level differences rather than a strictly matched frozen-backbone linear-probe comparison. All reported results use four seeds: 0042, 2024, 2025, and 3407.

The Pareto efficiency experiment fixes the backbone to DINOv2 and the model family to TBS sparse MIL with DepthAug, then sweeps $\rho \in \{0.1, 0.2, \dots, 1.0\}$ over four



FIGURE 3. Sample images from FruitsPark. The dataset contains seasonal and background variation within a park environment.



FIGURE 4. Sample images from UrbanStreetTree [5]. Street scenes contain complex backgrounds such as roads, buildings, vehicles, and neighboring trees.

seeds. We report accuracy, macro- F_1 , and average instances per bag. Patch instances are a direct proxy for encoder workload; wall-clock latency should be measured separately before making deployment-specific timing claims.

We also include two sparse-selection checks. First, a fixed-budget condition ablation compares full training/full evaluation, full training/sparse evaluation, random sparse training/evaluation, TBS sparse training/evaluation, and TBS sparse training with DepthAug followed by sparse RGB evaluation. Second, a ratio-wise comparison evaluates the proposed TBS-DepthAug configuration against random sparse sampling at the same bag ratios. These analyses are used to determine whether the proposed sampling and augmentation strategy improves over patch-count reduction alone.

C. IMPLEMENTATION DETAILS

All models are trained for 30 epochs. GAP baselines use batch size 32, whereas MIL models use batch size 1 because each image is represented as a patch bag. Dataset-specific normalization is computed from the training split. Sparse MIL and DepthAug models use softmax ABMIL attention, Sobel TBS scoring, $\lambda_a = 5 \times 10^{-4}$, no entropy regularization, and automatic mixed precision. The experiments were run on a workstation with NVIDIA RTX A6000 GPUs, CUDA 12.9, an AMD Ryzen Threadripper 7960X CPU, and 62 GiB system memory.

V. RESULTS

A. OVERALL COMPARISON

Table 2 reports the complete four-seed results for the fixed-budget backbone comparison. On UrbanStreetTree, sparse MIL improves over image-level ViT GAP, and DepthAug provides a further gain. ViT DepthAug achieves the best mean result in this fixed-budget comparison, 95.10% accuracy and 94.27% macro- F_1 . DINOv2 DepthAug is close, with 94.93% accuracy and 94.13% macro- F_1 . On FruitsPark, the strongest image-level models remain highly competitive; DINOv3 GAP reaches 97.62% accuracy and 97.60% macro- F_1 . The best sparse DepthAug model on FruitsPark is DINOv2 DepthAug, with 93.81% accuracy and 93.87% macro- F_1 .

B. EFFECT OF DEPTH-GUIDED AUGMENTATION

DepthAug is most effective on UrbanStreetTree. Relative to the corresponding sparse MIL model, paired-seed macro- F_1 improvements are +3.74 percentage points for ViT, +3.59 for DINOv2, +2.57 for DINOv3, and +4.28 for Swin. This supports the interpretation that training-time background perturbation is useful in street scenes where background shortcuts are severe.

On FruitsPark, the effect is mixed. DINOv2 and ViT gain from DepthAug, but DINOv1 and DINOv3 do not. This difference is important for interpretation: FruitsPark is a validation-only dataset collected in a specific park environment, and strong image-level foundation models may benefit from stable domain regularities. We therefore use FruitsPark

TABLE 2. Complete four-seed results for all configurations. Values are mean $\pm$ standard deviation in percent. FruitsPark is evaluated on validation; UrbanStreetTree is evaluated on test. Colored arrows show the change in percentage points relative to the reported Image-level GAP protocol of the same backbone family: blue upward arrows indicate gains and red downward arrows indicate losses.

Group	Model	FruitsPark		UrbanStreetTree	
		Accuracy	Macro- F_1	Accuracy	Macro- F_1
Image-level	ResNet18 GAP	93.93 $\pm$ 1.96	93.85 $\pm$ 2.10	83.92 $\pm$ 3.28	81.40 $\pm$ 3.95
Image-level+Aug	ResNet18 GAP Aug	92.14 $\pm$ 2.59 ($\downarrow$ 1.79)	92.01 $\pm$ 2.72 ($\downarrow$ 1.84)	83.57 $\pm$ 2.32 ($\downarrow$ 0.35)	81.55 $\pm$ 1.61 ($\uparrow$ 0.16)
Sparse MIL	ResNet18 MIL	35.36 $\pm$ 2.84 ($\downarrow$ 58.57)	33.97 $\pm$ 3.14 ($\downarrow$ 59.88)	9.79 $\pm$ 5.62 ($\downarrow$ 74.13)	6.96 $\pm$ 3.26 ($\downarrow$ 74.44)
Image-level	ViT GAP	95.12 $\pm$ 1.57	95.05 $\pm$ 1.62	84.62 $\pm$ 2.74	81.77 $\pm$ 3.12
Image-level+Aug	ViT GAP Aug	94.76 $\pm$ 1.23 ($\downarrow$ 0.36)	94.65 $\pm$ 1.44 ($\downarrow$ 0.41)	84.97 $\pm$ 3.81 ($\uparrow$ 0.35)	82.11 $\pm$ 4.22 ($\uparrow$ 0.34)
Sparse MIL	ViT MIL	86.67 $\pm$ 1.60 ($\downarrow$ 8.45)	86.42 $\pm$ 1.58 ($\downarrow$ 8.63)	91.61 $\pm$ 1.28 ($\uparrow$ 6.99)	90.53 $\pm$ 1.78 ($\uparrow$ 8.76)
Sparse DepthAug	ViT DepthAug	88.21 $\pm$ 2.18 ($\downarrow$ 6.90)	88.24 $\pm$ 2.21 ($\downarrow$ 6.81)	95.10 $\pm$ 1.81 ($\uparrow$10.49)	94.27 $\pm$ 2.37 ($\uparrow$12.50)
Image-level	DINOv1 GAP	60.24 $\pm$ 3.12	59.49 $\pm$ 3.16	57.17 $\pm$ 8.22	54.51 $\pm$ 8.23
Image-level+Aug	DINOv1 GAP Aug	49.29 $\pm$ 10.08 ($\downarrow$ 10.95)	47.34 $\pm$ 9.99 ($\downarrow$ 12.15)	49.30 $\pm$ 7.28 ($\downarrow$ 7.87)	45.33 $\pm$ 6.83 ($\downarrow$ 9.18)
Sparse MIL	DINOv1 MIL	90.71 $\pm$ 0.91 ($\uparrow$ 30.48)	90.55 $\pm$ 1.13 ($\uparrow$ 31.05)	92.83 $\pm$ 3.49 ($\uparrow$ 35.66)	91.16 $\pm$ 4.52 ($\uparrow$ 36.65)
Sparse DepthAug	DINOv1 DepthAug	88.57 $\pm$ 0.87 ($\uparrow$ 28.33)	88.37 $\pm$ 0.95 ($\uparrow$ 28.88)	94.06 $\pm$ 1.46 ($\uparrow$ 36.89)	92.87 $\pm$ 1.73 ($\uparrow$ 38.36)
Image-level	DINOv2 GAP	41.31 $\pm$ 3.70	39.50 $\pm$ 4.28	45.45 $\pm$ 3.33	42.58 $\pm$ 3.22
Image-level+Aug	DINOv2 GAP Aug	41.67 $\pm$ 0.99 ($\uparrow$ 0.36)	41.34 $\pm$ 1.35 ($\uparrow$ 1.84)	43.71 $\pm$ 3.26 ($\downarrow$ 1.75)	40.59 $\pm$ 3.14 ($\downarrow$ 1.99)
Sparse MIL	DINOv2 MIL	92.50 $\pm$ 1.67 ($\uparrow$ 51.19)	92.52 $\pm$ 1.66 ($\uparrow$ 53.02)	92.13 $\pm$ 2.16 ($\uparrow$ 46.68)	90.54 $\pm$ 2.65 ($\uparrow$ 47.96)
Sparse DepthAug	DINOv2 DepthAug	93.81 $\pm$ 2.37 ($\uparrow$ 52.50)	93.87 $\pm$ 2.39 ($\uparrow$ 54.37)	94.93 $\pm$ 2.31 ($\uparrow$ 49.48)	94.13 $\pm$ 2.74 ($\uparrow$ 51.55)
Image-level	DINOv3 GAP	97.62 $\pm$ 1.03	97.60 $\pm$ 1.04	86.71 $\pm$ 0.99	84.67 $\pm$ 0.89
Image-level+Aug	DINOv3 GAP Aug	95.24 $\pm$ 1.23 ($\downarrow$ 2.38)	95.25 $\pm$ 1.26 ($\downarrow$ 2.34)	87.94 $\pm$ 2.01 ($\uparrow$ 1.22)	86.88 $\pm$ 1.73 ($\uparrow$ 2.21)
Sparse MIL	DINOv3 MIL	83.33 $\pm$ 2.06 ($\downarrow$ 14.29)	83.15 $\pm$ 2.06 ($\downarrow$ 14.45)	88.46 $\pm$ 1.85 ($\uparrow$ 1.75)	87.76 $\pm$ 1.84 ($\uparrow$ 3.09)
Sparse DepthAug	DINOv3 DepthAug	86.55 $\pm$ 0.81 ($\downarrow$ 11.07)	86.54 $\pm$ 0.77 ($\downarrow$ 11.06)	91.26 $\pm$ 1.46 ($\uparrow$ 4.55)	90.33 $\pm$ 1.27 ($\uparrow$ 5.66)
Image-level	Swin GAP	97.14 $\pm$ 1.69	97.09 $\pm$ 1.74	89.34 $\pm$ 0.67	87.10 $\pm$ 1.22
Image-level+Aug	Swin GAP Aug	95.12 $\pm$ 1.06 ($\downarrow$ 2.02)	95.09 $\pm$ 1.04 ($\downarrow$ 2.00)	87.76 $\pm$ 3.54 ($\downarrow$ 1.57)	86.57 $\pm$ 3.64 ($\downarrow$ 0.53)
Sparse MIL	Swin MIL	80.60 $\pm$ 2.73 ($\downarrow$ 16.55)	80.51 $\pm$ 2.64 ($\downarrow$ 16.58)	89.69 $\pm$ 1.93 ($\uparrow$ 0.35)	87.61 $\pm$ 2.22 ($\uparrow$ 0.51)
Sparse DepthAug	Swin DepthAug	84.40 $\pm$ 0.81 ($\downarrow$ 12.74)	84.31 $\pm$ 0.90 ($\downarrow$ 12.78)	93.18 $\pm$ 1.55 ($\uparrow$ 3.85)	91.89 $\pm$ 2.08 ($\uparrow$ 4.79)

as a secondary validation of the method's behavior rather than as the sole basis for a superiority claim.

C. SPARSE-SELECTION ABLATION

The fixed-budget condition ablation in Fig. 7 compares whether sparse inference alone is sufficient and whether TBS and DepthAug improve over random sparse training. On FruitsPark, full training and full evaluation reach 88.6% accuracy in this protocol. Switching to full training with sparse evaluation reduces accuracy to 78.9%, showing that reducing the evaluation budget alone is not sufficient. Random sparse training/evaluation reaches 76.3%, TBS sparse training/evaluation reaches 78.2%, and TBS sparse with DepthAug improves to 85.0%. On UrbanStreetTree, full training/full evaluation reaches 94.2%, while full training/sparse evaluation, random sparse, TBS sparse, and TBS sparse with DepthAug reach 90.4%, 88.3%, 89.7%, and 91.3%, respectively. Thus, at a fixed sparse budget, TBS gives a modest gain over random sparse sampling, and DepthAug provides the largest additional gain. Full-patch processing remains a strong reference, but the gap is substantially reduced by combining sparse selection with training-time background perturbation.

D. PARETO EFFICIENCY OF SPARSE PATCH SELECTION

Table 3 summarizes the Pareto efficiency experiment. On FruitsPark, the best accuracy is obtained at $\rho = 1.0$, with 96.79% accuracy and 66.0 instances. The efficient point

within one percentage point of the best result is $\rho = 0.8$, which uses 53.0 instances and loses only 0.71 percentage points. On UrbanStreetTree, the best accuracy is obtained at $\rho = 0.9$, with 96.33% accuracy and 367.2 instances. The efficient point is $\rho = 0.4$, which reaches 95.98% accuracy and 95.22% macro- F_1 with only 163.3 instances, losing 0.35 percentage points while using about 44.5% of the instances used by the best-accuracy setting.

Fig. 8 compares the proposed TBS-DepthAug ratio sweep with random sparse sampling using mean and standard deviation over four seeds. Random sparse sampling is a strong stochastic baseline: because it can occasionally include highly informative patches, it can match or exceed the proposed method at isolated patch budgets. However, its performance is less stable across the ratio sweep, and unfavorable samples can produce noticeably weaker results. On FruitsPark, random sparse peaks at $\rho = 0.6$ and then degrades as more random patches are added, and on UrbanStreetTree it peaks at $\rho = 0.3$ before dropping at nearby ratios. The proposed method does not dominate random sampling at every budget, but it reaches the highest observed mean accuracy on both datasets and forms a more reliable high-performance region in the ratio sweep. This supports the interpretation that the contribution is not patch-count reduction alone, but more stable structure-biased sparse selection combined with depth-guided training-time perturbation.

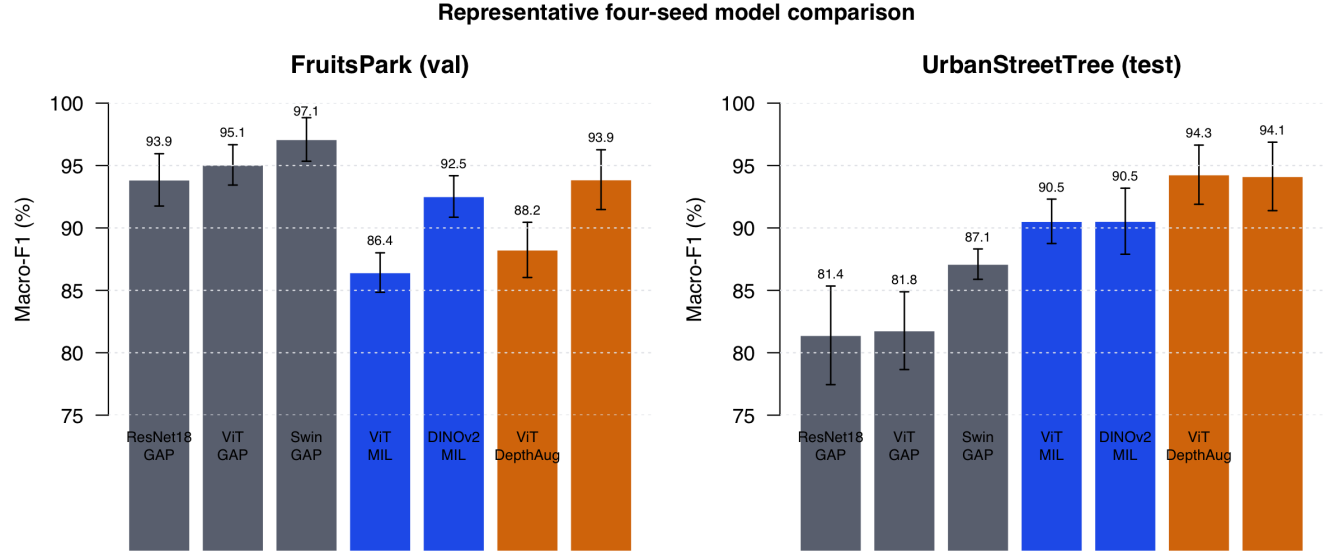


FIGURE 5. Representative four-seed macro- F_1 comparison. Gray bars indicate image-level GAP baselines, blue bars indicate sparse MIL, and orange bars indicate sparse MIL with DepthAug.

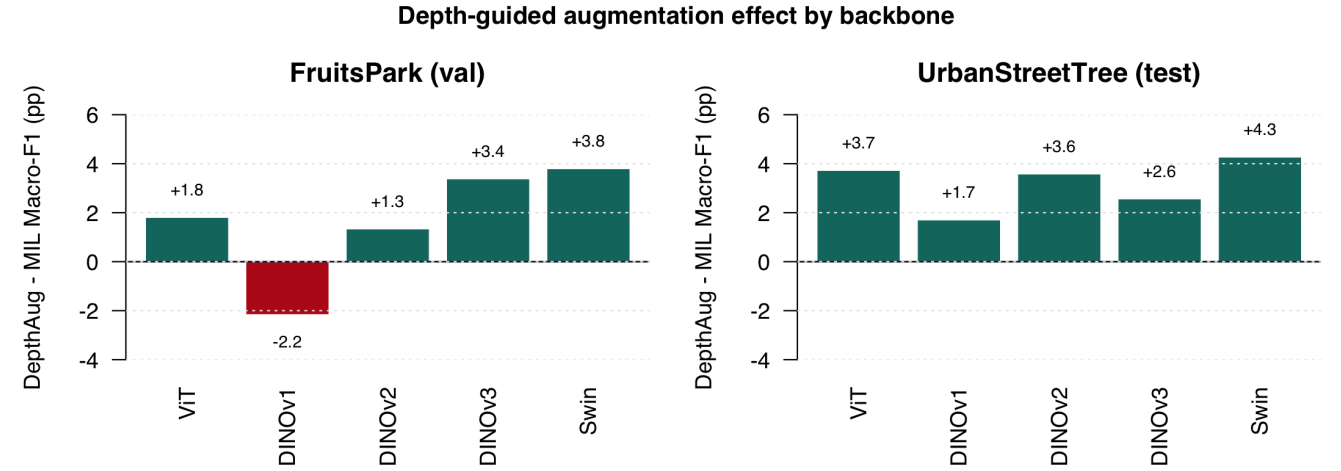


FIGURE 6. DepthAug delta relative to corresponding sparse MIL models. The effect is consistently positive on UrbanStreetTree and dataset-dependent on FruitsPark.

E. QUALITATIVE ATTENTION BEHAVIOR

Attention visualizations can be useful diagnostics for checking whether high-weight patches correspond to trunks, branches, or background regions. However, attention weights should not be treated as causal explanations by themselves [37]. For this reason, we use attention behavior only as supporting diagnostic evidence and base the main claims on the four-seed quantitative comparison and Pareto analysis. Selected-patch visualizations are most informative when generated from the exact TBS-DepthAug setting used in the quantitative experiments; visualizations from earlier internal variants are therefore not used as primary evidence.

VI. DISCUSSION

A. WHY SPARSE MIL HELPS

Mid-range tree classification requires the model to preserve fine local evidence while reasoning over the entire object structure. Standard image-level resizing can dilute trunk and bark information, especially when the tree occupies a small region. ABMIL avoids forcing the whole image into one global descriptor; it lets the model assign weights to local patches. TBS makes this practical by reducing the number of instances and biasing the bag toward structural patches before feature extraction. The random-comparison experiment also shows that random sparse sampling is not a weak baseline: by chance, it can select useful patches and become competitive at some ratios. The value of TBS-DepthAug is therefore not that it dominates random sampling at every patch budget, but

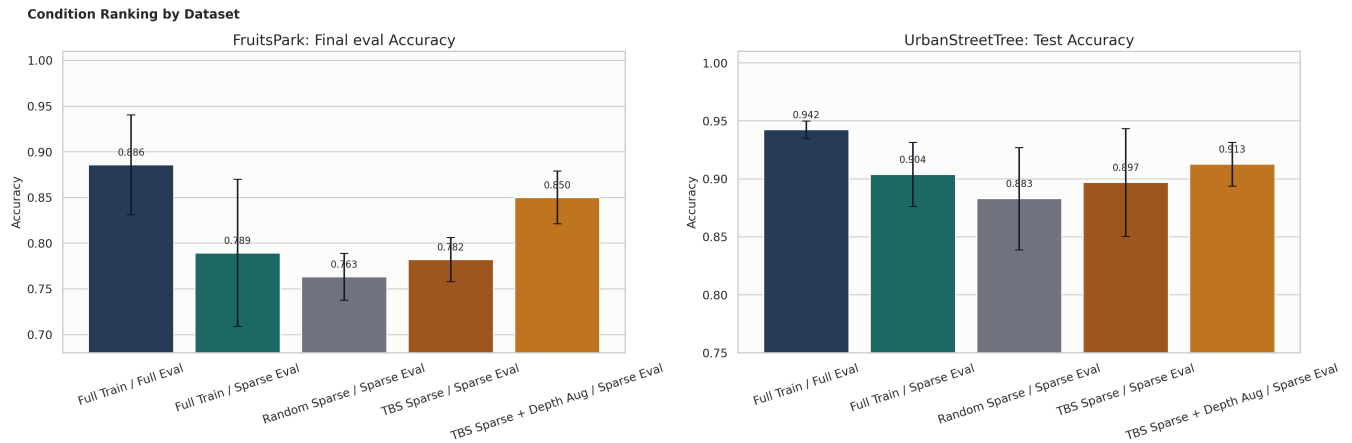


FIGURE 7. Fixed-budget condition ablation comparing full processing, sparse evaluation, random sparse training/evaluation, TBS sparse training/evaluation, and TBS sparse training with DepthAug.

TABLE 3. Pareto efficiency summary for TBS sparse MIL with DepthAug.

Dataset	Choice	Bag ratio	Accuracy (%)	Macro- F_1 (%)	Instances	Gap to best (pp)
FruitsPark	Best accuracy	1.0	96.79 ± 1.57	96.79	66.0	0.00
FruitsPark	Efficient within 1pp	0.8	96.07 ± 2.76	96.01	53.0	0.71
UrbanStreetTree	Best accuracy	0.9	96.33 ± 1.84	95.83	367.2	0.00
UrbanStreetTree	Efficient within 1pp	0.4	95.98 ± 0.88	95.22	163.3	0.35

that it reduces dependence on favorable random draws, improves the fixed sparse condition, and produces higher best-accuracy points while retaining a practical sparse operating point.

B. WHY DEPTHAUG HELPS ON URBANSTREETTREE

UrbanStreetTree contains complex street backgrounds and an independent test split, making it a strong setting for evaluating background shortcut suppression. The consistent DepthAug gains across backbones suggest that perturbing depth-derived background regions during training prevents the classifier from relying on scene context alone. Importantly, DepthAug does not impose a depth sensor requirement at inference, which is essential for mobile and low-cost deployment.

C. DATASET-DEPENDENT BEHAVIOR

FruitsPark shows a different pattern: DINOv3 GAP is the best overall method, and DepthAug does not improve all backbones. This does not invalidate the proposed framework. Instead, it indicates that the benefit of sparse depth-regularized MIL depends on dataset structure, split design, and background variability. Because FruitsPark has no independent test set, validation performance may reflect collection-domain regularities. Future work should add a held-out location or season split to measure generalization more rigorously.

D. EFFICIENCY INTERPRETATION

The Pareto results are central to the practical value of the method. In UrbanStreetTree, using 40% of candidate patches retains accuracy within 0.35 percentage points of the best setting while reducing average instances from 367.2 to 163.3. The random sparse baseline reaches a similar range, but its best point is slightly lower than the proposed method. This indicates that sparse patch processing itself is important, while TBS and DepthAug provide additional gains in the high-performance region. This does not yet equal a complete latency claim because preprocessing, patch scoring, GPU batching, and feature extraction overhead must be measured end-to-end. However, average instances are a direct and reproducible proxy for encoder workload, and the observed frontier shows that processing all patches is unnecessary for near-peak accuracy.

E. LIMITATIONS

This study has four main limitations. First, FruitsPark lacks an independent test set. Second, the current TBS score is a hand-designed Sobel measure; learnable or class-aware patch selection may improve robustness. Third, pseudo-depth maps can fail in cluttered vegetation or unusual lighting. Fourth, this paper reports patch-instance efficiency rather than a complete deployment latency benchmark. Future experiments should include held-out locations, end-to-end timing, confidence-aware depth augmentation, and direct comparisons with recent learned efficient MIL selectors.

Proposed vs Random: performance by bag ratio

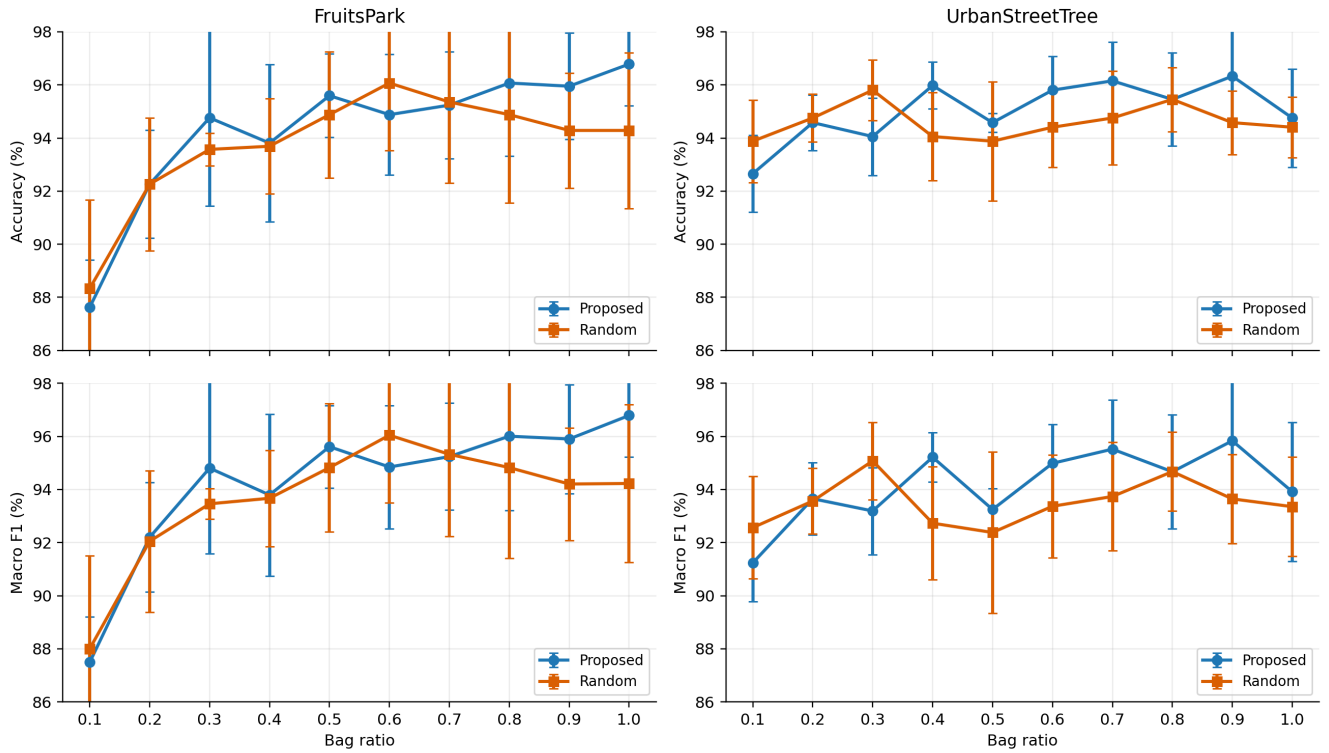


FIGURE 8. Proposed TBS-DepthAug versus random sparse sampling across bag ratios. Markers show four-seed means and error bars show standard deviations. Random sparse sampling is competitive at some isolated ratios, but its performance is more ratio-dependent; the proposed method gives the highest observed mean accuracy on both datasets while maintaining a more stable high-performance region.

VII. CONCLUSION

This paper presented a depth-regularized sparse MIL framework for mid-range tree species classification under background-dominant conditions. The method represents each image as a sparse bag of structurally informative patches, aggregates patch evidence with ABMIL, and uses depth-guided augmentation during training to reduce background shortcuts while preserving RGB-only inference. Fixed-budget four-seed experiments showed strong gains on UrbanStreetTree, where ViT DepthAug achieved 95.10% accuracy and 94.27% macro- F_1 . Fixed-budget ablations showed that TBS and DepthAug improve sparse training relative to random sparse sampling, while ratio-wise comparisons showed that random sparse sampling remains a strong baseline. Pareto analysis demonstrated that near-peak performance can be retained with fewer patch instances, especially on UrbanStreetTree, where a 0.4 bag ratio remained within 0.35 percentage points of the best result while using less than half as many instances. These findings support sparse, depth-regularized weakly supervised aggregation as a practical direction for ground-level tree recognition in complex real-world scenes.

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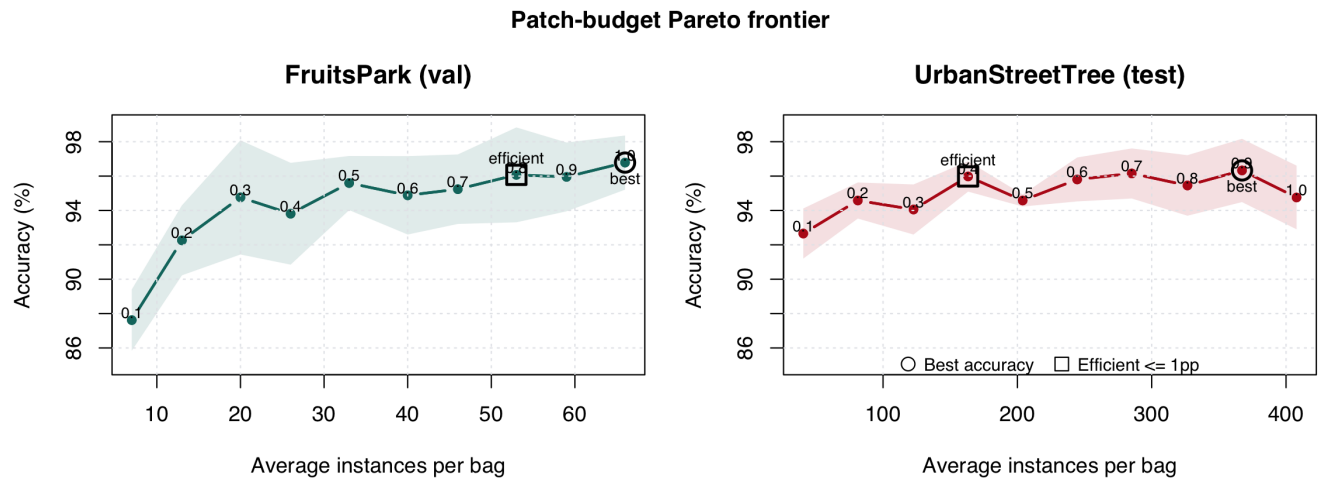


FIGURE 9. Patch-budget Pareto frontier for TBS sparse MIL with DepthAug. Circles mark the best-accuracy settings and squares mark the efficient settings within one percentage point of the best accuracy.

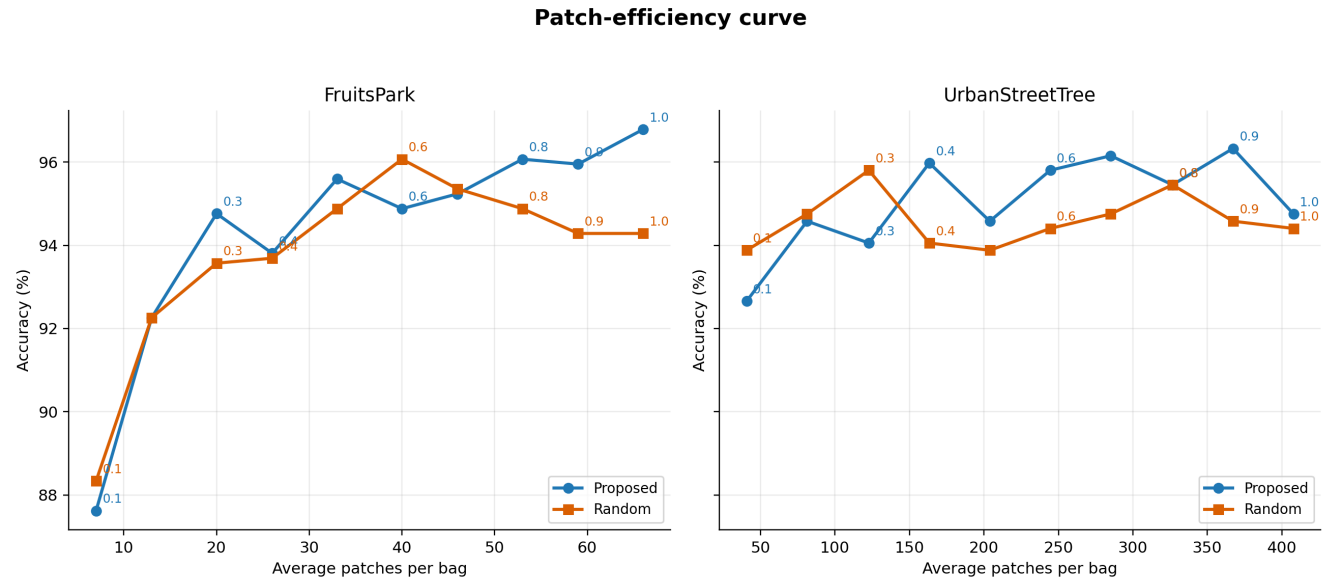


FIGURE 10. Patch-efficiency curves comparing the proposed TBS-DepthAug configuration with random sparse sampling across bag ratios.

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